# E-commerce Checkout Funnel Analysis – Case Study

---

**1. Problem Statement**

E-commerce platforms experience significant drop-offs across the user journey from product view to purchase. This leads to lost revenue opportunities and inefficient utilization of traffic.

The problem addressed in this project is:

Identifying where users drop off in the checkout funnel and understanding the underlying patterns affecting conversion.

---

**2. Objective**

- Analyze the checkout funnel stages
- Measure conversion and drop-off rates
- Identify high-impact problem areas
- Provide actionable recommendations to improve conversion and revenue

---

**3. Data Understanding**

The dataset consists of user-level interaction data with the following key fields:

- event_name → Defines user action (view, cart, checkout, purchase)
- user_pseudo_id → Unique user identifier
- device → Desktop, Mobile, Tablet
- country → User geography
- transaction_id → Purchase identifier
- purchase_revenue → Revenue generated

---

**4. Methodology**

**Step 1: Funnel Definition**

Defined funnel stages:

- Product View
- Add to Cart
- Checkout
- Purchase

---

**Step 2: Data Transformation**

- Cleaned raw data and standardized event names
- Mapped events to funnel stages
- Created stage order for correct funnel sequencing

---

**Step 3: Metric Calculation**

- Total Users per stage
- Conversion Rate = Purchase Users / Total Users
- Stage Conversion Rates (View→Cart, Cart→Checkout, etc.)
- Drop-off % between stages
- Average Order Value (AOV)
- Revenue and Revenue Loss

---

**Step 4: Segmentation Analysis**

- Device-level analysis
- Country-level analysis
- Funnel comparison across segments

---

**5. Key Findings**

**Funnel Performance**

- Major drop-off at **View → Cart stage (~87%)**
- Significant drop at **Cart → Checkout (~62%)**
- Checkout to Purchase conversion is relatively better but still suboptimal

---

**User Behaviour**

- Desktop users contribute highest conversions
- Mobile users form a large share but convert less
- Certain countries dominate traffic and revenue

---

**Revenue Analysis**

- Total revenue generated is limited relative to user volume
- High potential revenue loss due to low conversion rates
- AOV indicates moderate purchase value per user

---

## 6. Insights

- Users are not sufficiently motivated to add products to cart
- Checkout process introduces friction leading to abandonment
- Mobile experience is likely suboptimal
- Funnel inefficiencies directly impact revenue

---

## 7. Recommendations

### Product Page Optimization

- Improve product visuals and descriptions
- Add clear CTAs (Buy Now, Add to Cart)
- Display pricing and offers prominently

---

### Checkout Optimization

- Reduce number of steps
- Enable guest checkout
- Improve form usability

---

### Mobile Optimization

- Improve load time
- Simplify navigation
- Ensure responsive UI

---

### Retention & Remarketing

- Target users who abandon carts
- Offer incentives (discounts, reminders)

---

## 8. Conclusion

The analysis demonstrates that the primary issue lies in early-stage funnel drop-offs and checkout inefficiencies. Addressing these areas can significantly improve conversion rates and unlock revenue growth without increasing acquisition costs.